INSURE — Master Presentation & Defense Dossier

SYNAPSE '26 • TRACK: INSURTECH & IOT TELEMATICS • FULL SOLUTION SPECIFICATION

Team: **Techsquad**
Live Demo: **insurevsb.vercel.app**

1. OFFICIAL PROBLEM STATEMENT FULFILLMENT & INNOVATION CHARTER

Problem Statement: *"Create an insurance pricing platform that uses consented real-world behaviour data, such as vehicle telematics or approved IoT measurements, to inform premiums alongside insurer-approved risk factors."*

Challenges Stated in Brief:

- Traditional pricing ignores actual driving usage.
- Telematics data can be noisy, missing, or incomplete.
- Continuous tracking raises severe privacy concerns.
- Unexplained black-box premium changes destroy trust.

INSURE Complete Solution:

- **Consented IoT Stream:** ESP32 DevKit reads real pins (Speed, Helmet, Accel).
- **Data Integrity Gate:** Noise/dropouts never dock safety scores (Base: ₹2000).
- **Privacy Governance:** Dedicated DPDP consent portal; night driving excluded.
- **100% Explainable:** Linear deterministic rules with plain-text rupee audit logs.

2. THE HUMANIZED CONCEPT: RAVI VS. RAHUL

Ravi (The Safe Commuter)

Score: 100/100

Wears a helmet, cruises at 35–45 km/h, sober (0.00 ppm), smooth motion. In traditional insurance, he is unfairly overpaying to subsidize dangerous speeders.

INSURE Dynamic Rate: ₹1,800 / year (10.00% Dynamic Discount — Saves ₹200)

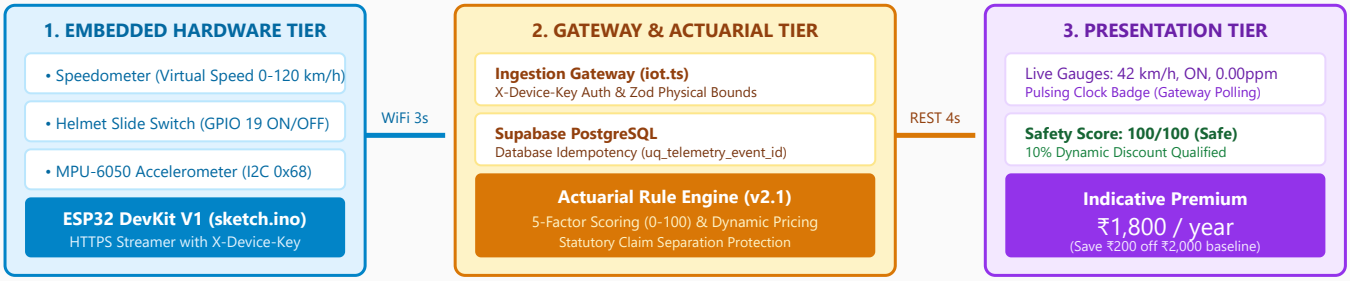
Rahul (The Risky Commuter)

Score: 42/100

Rides without a helmet (Pin 19 OFF), speeds at 95 km/h, brakes violently at the last second. Traditional flat-rate policies give him zero financial incentive to slow down.

INSURE Dynamic Rate: ₹2,400 / year (+20.00% Risk Surcharge — ₹400 penalty)

3. END-TO-END SYSTEM ARCHITECTURE DIAGRAM



4. THE 4 LIVE DEMONSTRATION SCENARIOS & INPUTS

Scenario A: Safe Commute (Ravi) → Score 100/100 → ₹1,800/yr

Inputs: Speed 38–42 km/h | Helmet: ON (Pin 19) | Alcohol: 0.00 ppm | Accel: 0.6 m/s² (1.0g).
Engine Action: Zero speeding or harsh braking → Safety Score: 100/100 → **10% Discount applied**.

Scenario B: Risky Highway Run (Rahul) → Score 42/100 → ₹2,400/yr

Inputs: Speed 95 km/h (Limit: 80) | Helmet: OFF | Harsh Decel: -5.2 m/s² | Alcohol: 0.28 ppm.
Engine Action: Speeding & braking penalties dock 58 pts → Score: 42/100 → **+20% Surcharge applied**.

Scenario C: Crash Impact Spike → Impact 7.42g → Fault: UNKNOWN

Inputs: Sudden impact shock: 7.42g (> 5.0g trigger) | Decel: -18.5 m/s² | Speed drops to 0.
Engine Action: Creates provisional incident in `accident_records`. **Fault is UNKNOWN (0% penalty)** pending human review on `/review`.

Scenario D: Two-Vehicle Correlated Crash (Bikes 001 & 002)

Inputs: Ather 450X and Ola S1 Pro collision.
Engine Action: Separate cryptographic event IDs recorded for both bikes without automated algorithmic blame.

5. ESP32 MICROCONTROLLER HARDWARE PINOUT & MAPPING

Sensor Component	ESP32 Pin	Physical Signal	Underwriting Telematics Function
MPU-6050 Accelerometer	I2C (SDA 21, SCL 22)	3-Axis Accel (m/s²)	Detects harsh braking (< -4 m/s²) and impact force (≥ 5.0g collision alert).
Helmet Slide Switch	GPIO 19 (Digital In)	HIGH (3.3V) / LOW (0V)	Detects helmet compliance (ON = Safe Commuter, OFF = Safety Breach).
Potentiometer (MQ-3)	GPIO 34 (ADC Analog)	0.00 – 0.80 ppm Alcohol	Simulates breathalyzer alcohol detection for zero-tolerance driving.
Speedometer Pulse	Firmware Internal	0.0 – 120.0 km/h	Tracks speeding breaches against highway limits (80 km/h).
Crash Pushbutton	GPIO 18 (Interrupt)	Impulse trigger	Forces high-g crash event (7.42g) to demonstrate neutral collision due process.

6. STATUTORY FAIRNESS: SEPARATION OF PRICING VS. CLAIMS SETTLEMENT



7. UNDERWRITING PROFITABILITY & PORTFOLIO ECONOMICS (PROFITABILITYSERVICE.TS)

Before INSURE (Traditional Flat-Rate Portfolio)

- Customers: 1,000 @ ₹2,000 flat premium = **₹20,00,000 Revenue**
- Claims Incurred: ₹12,00,000 | Operating Expenses: ₹5,00,000
- **Loss Ratio: 60.0%** | Expense Ratio: 25.0% | Combined Ratio: 85.0%
- **Net Underwriting Profit: ₹3,00,000.00**

With INSURE (Usage-Based Incentivized Portfolio)

- Customers: 1,300 @ ₹1,800 discounted premium = **₹23,40,000 Revenue**
- Claims Incurred: ₹13,00,000 | Total Expenses: ₹5,50,000
- **Loss Ratio: 55.6%** | Expense Ratio: 23.5% | Combined Ratio: 79.1%
- **Net Underwriting Profit: ₹4,90,000.00 (+63.3% Expansion!)**

8. 3-MINUTE LIVE HACKATHON DEMO SCRIPT (WHAT TO CLICK & SAY)

Time	Page / Action	What to Point to on Screen	Exact Words to Say to the Judges
0:00–0:45	Open `/simulator`	Status: `Live Stream Active`, Clock badge	"Judges, traditional motor insurance charges safe and reckless drivers the same flat rate. INSURE connects vehicles live to an ESP32 microcontroller to reward safe driving."
0:45–1:30	Click "*Start Safe Drive*"	Speed 42 km/h, Helmet ON, Score 100/100, ₹1,800/yr	"Our Ather 450X is cruising at 42 km/h with helmet ON. Our engine awards 100/100, dropping annual premium by 10% to ₹1,800."
1:30–2:15	Click "*Trigger Collision*"	Red Alert banner: 7.42g, Fault: UNKNOWN	"A 7.4g impact occurs! Unlike unfair black boxes, we never assume fault. The incident is logged as UNKNOWN for fair adjuster review on /review."
2:15–3:00	Open `/profitability`	Loss Ratio 55.6%, Net Profit: ₹4.90 Lakhs	"By encouraging safer driving, claims drop, expanding insurer underwriting profit from ₹3.0 Lakhs to ₹4.9 Lakhs. Fairness and profitability align perfectly."

9. ROLE-BASED NAVIGATION & ACCESS CONTROL (RBAC)

Policyholder (Driver):

- /dashboard (Personal rate)
- /trips (Trip breadcrumbs)
- /simulator (IoT live view)
- /consent (Privacy controls)

Commercial Underwriter:

- /admin (Fleet monitoring)
- /profitability (Loss ratios)
- /trips (Fleet telematics)
- /quotes (Pricing tables)

Compliance Officer:

- /compliance (Fairness view)
- /audit (Immutable logs)
- /consent (DPDP governance)
- /assessments (Rules audit)

Claims Adjuster:

- /review (Collision desk)
- Assigns verified fault
- Verifies claim payouts
- Neutrality enforcement